Rafi Mahmud Tumon

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Education

Technical University Berlin — Berlin, Germany
M.Sc. in Geodesy and Geoinformation Science; Specialization: Geoinformation Science — *October 2021 – Present*

Key Coursework: Photogrammetric Computer Vision, Geoinformatics, Geodatabases and Infrastructures, Space Geodesy, Adjustment Theory, Deep Learning for Geographic Data

Rajshahi University of Engineering and Technology — Rajshahi, Bangladesh
B.Sc. in Urban and Regional Planning — *March 2014 – November 2018*

Skills

Programming Languages: Python, JavaScript, MATLAB, SQL

Technologies: GeoPandas, Rasterio, OSMnx, Shapely, geemap, SAHI, Matplotlib, Seaborn, PyTorch, PyTorch Lightning, Scikit-learn, GDAL, PostgreSQL/PostGIS, Google Earth Engine

Tools: QGIS, ArcGIS Pro, ArcGIS Online, VSCode, Git, Jupyter Notebook, Docker

GIS & Remote Sensing: Spatial Data Management, Coordinate Reference Systems, Map Creation and Cartography, Spatial Joins, Processing Modeler, Workflow Automation, Change Detection, Land Use and Land Cover Classification, Vegetation Indices (NDVI, EVI, SAVI)

Machine Learning & Deep Learning: CNNs, Vision Transformers (ViT), Object Detection (YOLO), Multimodal Fusion, Model Evaluation & Classification

Work Experience

German Aerospace Center (DLR) — Berlin, Germany
Student Assistant — *January 2026 – Present*

- Developing a deep learning fusion model for urban tree health monitoring using aerial stereo imagery.
- Evaluating and comparing encoder architectures and fusion strategies to determine optimal configurations.
- Conducting data quality checks and geospatial validation using QGIS and Python to ensure dataset integrity.

OXG Glasfaser GmbH — Berlin, Germany
Working Student, Strategic Area Development — *June 2024 – November 2025*

- Improved operational efficiency by 20% through automated geospatial analysis using QGIS and Python to support fiber network deployment.
- Performed cost-benefit analyses and site assessments to identify optimal deployment areas while navigating geographical constraints and market competition.
- Developed Python-based geocoding tool for German addresses, reducing processing time by 30%.

Department of Civil Engineering, RUET — Rajshahi, Bangladesh
Project Assistant (GIS and Geospatial Planner) — *February 2020 – April 2020*

- Contributed to an airport master planning project using GIS analysis to support regional connectivity for underserved communities in northwestern Bangladesh.
- Collaborated with engineering teams and stakeholders to integrate spatial analysis into transportation planning decisions.

Urban Development Directorate — Rajshahi, Bangladesh
GIS Analyst — *May 2018 – July 2018*

- Conducted socio-economic surveys and spatial analysis for an urban planning project in Bagha Upazilla, collecting field data from local communities.
- Maintained geodatabases tracking urbanization indicators to support resource allocation decisions for deprived areas.

Awards & Achievements

Planning and Design of Compact Township, 2nd Runner Up: Inter-University Undergraduate Planning Students Competition, Bangladesh University of Engineering and Technology (BUET).

Publication: "Identification of Determinant Factors Influencing Modal Choice Behavior and Satisfaction Level of Commuters: A Case of Rajshahi City." *Trends in Civil Engineering and its Architecture*, Lupine Publishers. DOI: 10.32474/TCEIA.2019.03.000171

Projects

Multi-view Deep Learning for Urban Tree Crown Condition Assessment | *Master's Thesis* Present

- Implementing and comparing multiple YOLO architectures for tree crown detection on large-format GeoTIFF imagery, with YOLOv12l achieving the best performance at an F1 score of 0.73.
- Constructing geospatial preprocessing pipelines for spatial joins, CRS-consistent raster cropping, and multi-channel patch extraction.
- Designing a PyTorch Lightning classification framework for tree vitality assessment.

Flood Vulnerability Assessment Using Geospatial Analysis | *Bachelor's Thesis* November 2018

- Conducted vulnerability assessment for three flood-prone districts in Bangladesh using satellite imagery to identify at-risk communities.
- Developed geospatial methodologies and produced flood risk maps to support disaster planning.

Energy Consumption Analysis and Feature Engineering | *GitHub*

- Analyzed the UCI Appliances Energy Prediction dataset, resampling 10-minute sensor readings to hourly and consolidating 18 sensors into 7 aggregates.
- Built a preprocessing pipeline with cyclical time encoding, lag/rolling features, and leakage-safe chronological train-val-test splits.

End-to-End Geospatial Data Pipeline | *GitHub*

- Built a geospatial data pipeline using USGS DEM, LiDAR, OpenStreetMap, and Sentinel-2 satellite imagery as data sources.
- Automated raster processing, spatial analysis, and PostGIS database loading across 8 Python scripts with Docker for the database.
- Performed spatial queries including road buffering, building height extraction from LiDAR, building road access, and land suitability scoring inside PostGIS.

Weather Analytics Dashboard | *GitHub* | *Live App*

- Built an ETL pipeline that fetches daily weather data for global megacities from the Open-Meteo API and loads it into a PostgreSQL database on Supabase.
- Scheduled daily runs with GitHub Actions as part of a CI/CD pipeline and developed pytest unit tests for the transform layer.
- Built a multi-page Streamlit dashboard with temperature trends, monthly heatmaps, year-over-year comparisons, and anomaly detection using z-scores.

Forest Loss Analysis in Colombia's Indigenous Lands | *GitHub*

- Analyzed 10 years of satellite images using Google Earth Engine and Python to track deforestation and its effects on vulnerable communities, revealing a 15% loss in tree cover.

Wildfire Data Analysis | *GitHub*

- Developed a Python script to analyze wildfire patterns in 8 countries (analyzing 1,234 fires total) to help with disaster planning.

Berlin's Environmental Dynamics Analysis | *GitHub*

- Assessed Urban Heat Island effects using satellite data, finding that built-up areas experienced the highest temperatures compared to other land types.

Languages

English (Professional), **German** (Elementary - A2), **Bengali** (Native)